Model: 1-layer ACR - GNN.

Input graph:

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{Feature matrix} \qquad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{Adjacency matrix.}$$

Local aggregation:

$$\begin{array}{c} e0\_0 \\ e1\_0 \end{array} \underset{2 \times 2}{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}} \begin{array}{c} e0\_1 \\ e1\_1 \end{array} \cdot \begin{array}{cc} & \begin{array}{ccc} \_0 & \_1 & \_2 \end{array} \\ \begin{array}{c} x0 \\ x1 \end{array} & \underset{2 \times 3}{\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}} \end{array} =$$

$$\begin{array}{c} x2 \\ = \\ x3 \end{array} \begin{bmatrix} \overset{\_0}{e0\_0 \cdot x0\_0 + e0\_1 \cdot x1\_0} & \overset{\_1}{e0\_0 \cdot x0\_1 + e0\_1 \cdot x1\_1} & \overset{\_2}{e0\_0 \cdot x0\_2 + e0\_1 \; x1\_2} \\ e1\_0 \cdot x0\_0 + e1\_1 \cdot x1\_0 & e1\_0 \cdot x0\_1 + e1\_1 \cdot x1\_1 & e1\_0 \cdot x0\_2 + e1\_1 \; x1\_2 \end{bmatrix} = \begin{array}{cc} & \begin{array}{ccc} \_0 & \_1 & \_2 \end{array} \\ \begin{array}{c} x2 \\ x3 \end{array} & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

Global aggregation:

$$\begin{array}{cc} & \begin{array}{ccc} \_0 & \_1 & \_2 \end{array} \\ \begin{array}{c} X0 \\ X1 \end{array} & \underset{2 \times 3}{\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}} \end{array} \rightsquigarrow \begin{array}{c} x4 \\ x5 \end{array} \begin{array}{ccc} X0\_0 + X1\_0 & X0\_1 + X1\_1 & X0\_2 + X1\_2 \\ X0\_0 + X1\_0 & X0\_1 + X1\_1 & X0\_2 + X1\_2 \end{array}$$

this should be dublicated # nodes.

$$\underset{(2,3)}{x} \qquad\qquad\qquad\qquad \underset{(3,\,3)}{V}$$

$$\begin{array}{c c} X_0 & \multirow{2}{*}{} \\ X_1 & \end{array}
\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} -0.0230 & -0.4249 & -0.5190 \\ 0.1664 & -0.2224 & 0.5175 \\ 0.0743 & 0.1548 & 0.3314 \end{bmatrix} +$$

$$\_0 \;\; \_1 \;\; \_2$$

$$\underset{(3,)}{b}$$

$$+ \begin{bmatrix} 0.3200 & -0.1745 & 0.2483 \\ 0.3200 & -0.1745 & 0.2483 \end{bmatrix} =$$

$$xV^T$$

$$= \begin{array}{c} X6 \\ X7 \end{array} \begin{bmatrix} 0.0743 & 0.1548 & 0.3314 \\ -0.0230 & -0.4249 & -0.5190 \end{bmatrix} +$$

$$b$$

$$+ \begin{bmatrix} 0.3200 & -0.1745 & 0.2483 \\ 0.3200 & -0.1745 & 0.2483 \end{bmatrix} =$$

$$= \begin{array}{c} X8 \\ X9 \end{array} \begin{bmatrix} 0.3943 & -0.0197 & 0.5797 \\ 0.297 & -0.5994 & -0.2707 \end{bmatrix} \qquad x\, V^T + b_V \; !$$

agg_Local (over $\mathcal{N}(v)$)

$$\underset{(3\times 3)}{A}$$

$$\begin{array}{c} X2 \\ X3 \end{array} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} -0.9673 & 0.0214 & 0.3998 \\ -0.1476 & 0.2282 & -0.3737 \\ -0.1174 & 0.3464 & -0.2479 \end{bmatrix} +$$

$$+ \begin{bmatrix} 0.6466 & -0.1188 & 0.7938 \\ 0.6466 & -0.1188 & 0.7938 \end{bmatrix} =$$

agg-local A^T

$$= \begin{array}{c} X10 \\ X11 \end{array} \begin{bmatrix} -0.9673 & 0.0214 & 0.3998 \\ -0.1174 & 0.3464 & -0.2479 \end{bmatrix} +$$

$$+ \begin{bmatrix} 0.6466 & -0.1188 & 0.7938 \\ 0.6466 & -0.1188 & 0.7938 \end{bmatrix} =$$

$$= \begin{array}{c} X12 \\ X13 \end{array} \begin{bmatrix} -0.3207 & -0.0974 & 1.1936 \\ 0.5292 & 0.2276 & 0.5459 \end{bmatrix} \qquad \text{agg-local } A^T + b_A$$

agg-global (over G).

$$\underset{(2,3)}{Z} \qquad \underset{(3,3)}{R}$$

$$\begin{array}{c} X4 \\ X5 \end{array} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0.0741 & -0.5356 & 0.1367 \\ 0.0609 & -0.3635 & -0.0216 \\ 0.0611 & -0.1462 & -0.0530 \end{bmatrix} +$$

$$+ \begin{bmatrix} -0.0986 & -0.4034 & -0.1789 \\ -0.0986 & -0.4034 & -0.1789 \end{bmatrix} =$$

agg-global R^T

$$= \begin{array}{c} X14 \\ X15 \end{array} \begin{bmatrix} 0.1352 & -0.6818 & 0.0837 \\ 0.1352 & -0.6818 & 0.0837 \end{bmatrix} +$$

$$+ \begin{bmatrix} -0.0986 & -0.4034 & -0.1789 \\ -0.0986 & -0.4034 & -0.1789 \end{bmatrix} =$$

$$= \begin{array}{c} X16 \\ X17 \end{array} \begin{bmatrix} 0.0366 & -1.0852 & -0.0952 \\ 0.0366 & -1.0852 & -0.0952 \end{bmatrix} \qquad \text{agg-global } R^T + b_R$$

$$\left(x\, V^T + b_V \right) + \left(agg\_local\, A^T + b_A \right) + \left(agg\_global\, R^T + b_R \right)$$

$$\begin{array}{c} x_8 \\ x_9 \end{array} \begin{bmatrix} 0.3943 & -0.0197 & 0.5797 \\ 0.297 & -0.5994 & -0.2707 \end{bmatrix} + \qquad (2, 3)$$

$$+ \begin{array}{c} x_{12} \\ x_{13} \end{array} \begin{bmatrix} -0.3207 & -0.0974 & 1.1936 \\ 0.5292 & 0.2276 & 0.5459 \end{bmatrix} + \qquad (2, 3)$$

$$+ \begin{array}{c} x_{16} \\ x_{17} \end{array} \begin{bmatrix} 0.0366 & -1.0852 & -0.0952 \\ 0.0366 & -1.0852 & -0.0952 \end{bmatrix} = \qquad (2, 3)$$

$$= \begin{array}{c} x_{18} \\ x_{19} \end{array} \begin{bmatrix} 0.1102 & -1.2023 & 1.6781 \\ 0.8628 & -1.457 & 0.18 \end{bmatrix} \qquad \underline{comb}.$$

Apply activation function : ReLU.

$$ReLU\left(\begin{array}{c} x_{18} \\ x_{19} \end{array} \begin{bmatrix} 0.1102 & -1.2023 & 1.6781 \\ 0.8628 & -1.457 & 0.18 \end{bmatrix} \right) =$$

$$= \begin{array}{c} x_{20} \\ x_{21} \end{array} \begin{bmatrix} 0.1102 & 0 & 1.6781 \\ 0.8628 & 0 & 0.18 \end{bmatrix} \qquad \underline{act}. \quad !$$

Applying Batch Normalization.

$$act[i][j] \cdot a[j] + c[j]$$

selected node ↗ ↖ feature / hidden dimension.

$$
\begin{array}{c}
X_{20} \\ X_{21}
\end{array}
\begin{bmatrix} 0.1102 & 0 & 1.6781 \\ 0.8628 & 0 & 0.18 \end{bmatrix}
\cdot
\begin{bmatrix} 4.837038 & 0 & 0 \\ 0 & 316 & 0 \\ 0 & 0 & 2.691807 \end{bmatrix}
$$

2×3

$$
+ \begin{bmatrix} -2.2046876 & 0.20484535 & -2.4879744 \\ -2.2046876 & 0.20484535 & -2.4879744 \end{bmatrix} =
$$

$$
= \begin{bmatrix} 0.5330 & 0 & 4.528868 \\ 4.17339 & 0 & 0.485785 \end{bmatrix} +
$$

$$
+ \begin{bmatrix} -2.2046876 & 0.20484535 & -2.4879744 \\ -2.2046876 & 0.20484535 & -2.4879744 \end{bmatrix} =
$$

$$
\begin{array}{c}
X_{22} \\ X_{23}
\end{array}
\begin{bmatrix} -1.6716 & 0.20484535 & 2.0406 \\ 1.9686 & 0.20484555 & -2.0022 \end{bmatrix} BN(act)
$$

Layer is finished.

Linear Pregiction is started.

$$BN(act) \cdot W_{LP}^{T} + b\_LP$$

$$\begin{array}{c} x22 \\ x23 \end{array}\begin{bmatrix} -1.6716 & 0.20484535 & 2.0406 \\ 1.9686 & 0.20484535 & -2.0022 \end{bmatrix} \cdot$$

$$\underset{3\times 2}{\begin{bmatrix} -1.1680429 & 1.112733 \\ 0.7259423 & 0.0068289 \\ 0.9701 0756 & -1.0101824 \end{bmatrix}} + \underset{(3,)}{\begin{bmatrix} 0.32715797 & -0.44940925 \\ 0.32715797 & -0.44940925 \end{bmatrix}} =$$

2×3

$$= \begin{array}{c} x24 \\ x25 \end{array}\begin{bmatrix} 4.0807 & -3.9200 \\ -4.0930 & 4.2145 \end{bmatrix}$$

$$+ \begin{bmatrix} 0.32715797 & -0.44940925 \\ 0.32715797 & -0.44940925 \end{bmatrix} =$$

$$= \begin{array}{c} x26 \\ x27 \end{array}\begin{bmatrix} 4.4079 & -4.3694 \\ -3.7658 & 3.7650 \end{bmatrix}$$

output?

Binary Classification: assigning classes

agmax (output, axis =1)

Int?

$$\begin{array}{c} x28 \\ x29 \end{array}\begin{bmatrix} \text{argmax}(4.4079, -4.3694) \\ \text{argmax}(-3.7658, 3.7650) \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Important: data type for x28, x29 has to be Int, not Float32.